

The Measurement of Diversity in Different Types of Biological Collections

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Information content may be used as a measure of the diversity of a many-species biological collection. The diversity of small collections, all of whose members can be identified and counted, is defined by Brillouin's measure of information. With larger collections it becomes necessary to estimate diversity; what is estimated is Shannon's measure of information which is a function of the population proportions of the several species. Different methods of estimation are appropriate for different types of collections. If the collection can be randomly sampled and the total number of species is known, Basharin's formula may be used. With a random sample from a population containing an unknown number of species, Good's method is sometimes applicable. With a patchy population of sessile organisms, such as a plant community, random samples are unobtainable since the contents of a randomly placed quadrat are not a random sample of the parent population. To estimate the diversity of such a community a method is proposed whereby the sample size is progressively increased by addition of new quadrats; as this is done the diversity of the pooled sample increases and then levels off. The mean increment in total diversity that results from enlarging the sample still more then provides an estimate of the diversity per individual in the whole population.

1. Introduction

The use of "information content" as a measure of the diversity of many-species biological collections is being increasingly adopted by ecologists (see, for example, Margalef, 1958; Hairston, 1959; MacArthur & MacArthur, 1961; MacArthur, 1964, 1965; Lloyd, 1964; Lloyd & Ghelardi, 1964). Diversity in this connexion means the degree of uncertainty attached to the specific identity of any randomly selected individual. The greater the number of species and the more nearly equal their proportions, the greater the uncertainty and hence the diversity. A collection may consist of organisms that naturally occur together (e.g. the sedges in a marsh, pairs of breeding birds in a forest, the corals of a reef); or else it may be an assemblage of organisms gathered by an ecologist that is to be studied as a unit (e.g. the

plankton in a water sample, a catch of insects from a light trap). For all such collections the information content as defined by Shannon & Weaver (1963) or by Brillouin (1960) provides a convenient measure of diversity. However, with so many dissimilar kinds of collections, it is impossible to prescribe a single method of determining diversity. The method must be adapted to the type of collection at hand.

To this end it is convenient to classify collections into five types, for each of which a different method is appropriate. This classification is given in section 2, together with short notes on the determination of diversity for four of the types. In section 3 a method of dealing with the fifth type is proposed and an example is given. Measurement of the "evenness" component of diversity (the ratio of the observed diversity to the maximum possible in a collection having the same number of species) is considered in section 4. Section 5 gives a glossary of the symbols used in the paper.

2. The Five Types of Collection

The five types of collection are defined by the following key:

(A) COLLECTIONS SMALL ENOUGH FOR ALL MEMBERS TO BE IDENTIFIED AND COUNTED (TYPE A);

(B) LARGER COLLECTIONS:

(a) *Collections from which a random sample can be drawn:*

(i) *Number of species known (Type B),*

(ii) *Number of species unknown;*

(A) *Sample gives a smooth species-abundance curve (Type C),*

(B) *Otherwise (Type D);*

(b) *Collections from which a random sample cannot be drawn (Type E).*

Type A

When a collection is not too large for all its members to be identified and counted, its diversity is given by Brillouin's formula:

$$H = \frac{1}{N} \log \frac{N!}{N_1! N_2! \dots N_s!}, \quad (1)$$

where N is the total number of individuals, s the number of species, and N_i the number of individuals in the i th species, so that $\sum_1^s N_i = N$. This is the diversity per individual. The total diversity (or total information content) is $HN = B$. In this paper, "diversity" by itself will always mean diversity per individual.

The base of the logarithms is at choice and affects only the size of the unit. With logarithms to base 2 the unit is a "binary digit" or "bit"; if the base is 10, the unit has been variously called a "bel" (Good, 1950), a "decimal digit" (Good, 1953) or a "decit" (Pielou, 1966a); if natural logarithms are used the unit may be called a "natural bel" (Good, 1950). In this paper, the information units will usually be left unspecified.

The diversity as defined by H is, of course, a population value, not an estimate from a sample subject to sampling variation. It is not assumed that the collection is a random sample from a larger population and such an assumption would usually be unjustified. Consider, for instance, a collection of insects caught in a light trap; it is impossible to specify precisely the boundaries of the region from which the insects came, and they are certainly not a random sample from the population of all the insects in any region. The catch should therefore be regarded as being itself a population, rather than as a sample of something bigger.

So far it has been assumed that the organisms in a collection are discrete individuals that can be counted. When the organisms are vegetatively-reproducing plants, this cannot be done. It then becomes necessary to measure the amount of each species in the collection (the contents of a single plant quadrat, for instance) in terms of units of some quantity such as weight, volume, or area of ground covered, and the units are then treated as "individuals", i.e. the number of individuals of the i th species, N_i , is the amount of the species in whatever units (e.g. grams) have been chosen for measurement.

Type B

Even though we always regard a collection as a population and not a sample, the collection itself may be too big for all its members to be identified and counted. Its diversity must then be estimated by taking a sample from the collection. Suppose it be possible to take a truly random sample. This would be so if the collection consisted of all the plankton organisms in a large volume of sea water, for example. The water could be thoroughly stirred and a volume then taken from it such that identification of all the contained organisms (which would be a random sample from the whole collection) was feasible. We also suppose that s , the total number of species in the population (as we shall now describe the whole collection), is known, regardless of whether all are represented in the sample.

Instead of determining the true population diversity H (as given by equation (1)), which can only be done for Type *A* collections, we now wish to estimate the average diversity from a sample. This average diversity, H' , is given by Shannon & Weaver's (1963) formula

$$H' = - \sum_{i=1}^s p_i \log p_i, \quad (2)$$

where p_i is the proportion of the i th species in the population. As Goldman (1953) has emphasized, Shannon's H' is only defined for average conditions in a large collection. The formula of Shannon & Weaver (2) is therefore inappropriate for Type *A* collections.

An estimate $\hat{H}'$ of H' may be obtained by putting

$$\hat{H}' = - \sum_{i=1}^s \hat{p}_i \log \hat{p}_i, \quad (3)$$

where $\hat{p}_i$ is the proportion of the i th species in the sample. However, as Basharin (1959) has shown, $\hat{H}'$ is a biased estimator of H' and if this bias is to be allowed for we must know s since the magnitude of the bias depends on it. When the units are natural bels (i.e. when natural logarithms are used in equations (2) and (3)), Basharin proves that the expectation of $\hat{H}'$ is

$$E(\hat{H}') = H' - (s-1)/2N, \quad (4)$$

where N is the size of the sample.

Thus, given a Type *B* collection (i.e. one for which s is known) we may estimate the bias that results from using the sample estimates $\hat{p}_i$ in place of the population values p_i and correct our estimate of H' accordingly.

Basharin also shows that $\hat{H}'$ is consistent and asymptotically normal. An estimator of its variance in large samples is given by

$$\widehat{\text{var}}(\hat{H}') = \frac{1}{N} \left[\sum_{i=1}^s \hat{p}_i \ln^2 \hat{p}_i - \hat{H}'^2 \right],$$

again using natural logarithms as in (4).

Type C

Unless all the members of a collection are identified the exact value of s is usually unknown. If it seems likely that the number of species in the sample falls only slightly short of s , no appreciable error will arise if a large sample from a Type *C* collection is treated as if it were from Type *B*. Often, however, a sample is found to contain many species represented by only one or two individuals. It is, then, reasonable to suppose that the whole collection contains numerous additional species besides those in the sample and the true value of s cannot be guessed, even approximately. With such samples, Good's (1953) method of estimating H' for the population may be used, provided that the observed species-abundance curve lends itself to acceptable smoothing. A species-abundance curve is drawn by plotting v_r , the number of species with r individuals on the ordinate, against successive values of r (1, 2, ...) on the abscissa. After smoothing this curve (see Good (1953) for

methods) smoothed values v'_r may be obtained in place of the observed v_r . Good then shows that an estimator of H' (in natural bels) is given by

$$\hat{H}' = \ln N - \frac{1}{N} \sum_r r v_r \left(g_r + \frac{d}{dr} \ln v'_r \right), \quad (5)$$

where

$$g_r = \sum_{j=1}^r \frac{1}{j} - \gamma$$

and $\gamma = 0.5772 \dots$ is Euler's constant.

Type D

If the species-abundance curve cannot be satisfactorily smoothed, the v'_r values required in equation (5) are unobtainable. It may prove possible to estimate the diversity of such a collection by the same method as that described for Type E collections (see below). I have not so far attempted this with an actual collection.

Type E

If a collection can be brought into the laboratory for study, a random sample can usually be drawn from it. But this can seldom be done when the collection is a natural community of sessile organisms, that is to be examined *in situ*, and whose spatial pattern is patchy. With discrete organisms one might obtain a random sample by sampling at randomly located points, but if the collection consists of patches of vegetatively-reproducing plants whose quantity is to be recorded by weight or volume even this is impossible. Such a collection is designated Type E.

Consider, for instance, a large tract of herbaceous vegetation that is sampled with randomly placed quadrats. By measuring the amount of each species in any one quadrat we may calculate H , as in equation (1), and obtain the diversity in that particular quadrat. We may also determine $\bar{H}$, the mean of H for a number of quadrats. Obviously, however, because of the patchiness of vegetation, $\bar{H}$ will always be less than H_{pop} , the diversity in the population as a whole. The coarser the vegetation pattern (relative to the size of quadrat used) the more markedly will $\bar{H}$ fall short of H_{pop} .

Suppose now that a total of z quadrats has been examined and that we take them in random order. Let $\mathcal{H}_1$ be the diversity of the first. Add the data from the second quadrat to those of the first and recalculate the diversity to obtain $\mathcal{H}_2$, the diversity of the combined pair of quadrats. Continuing thus we obtain a sequence of values $\mathcal{H}_1, \mathcal{H}_2, \dots, \mathcal{H}_k, \dots, \mathcal{H}_z$, which are the diversities of successively larger samples from the population. That is, $\mathcal{H}_k$ is the diversity of the combined contents of the first k of the randomly

placed quadrats. If the process were continued until the whole population had been sampled, obviously the final $\mathcal{H}$ would be identical with H_{pop} . If we plot $\mathcal{H}_k$ against k for $k = 1, 2, \dots, z$ we should expect $\mathcal{H}_k$ to increase (not necessarily monotonically) at first and then, provided a large enough z had been chosen, to level off.

The reason for predicting that the $\mathcal{H}_k$ versus k curve will level off is as follows. Once $\mathcal{H}_k$ has reached a certain level, addition of new quadrats to the sample will have two opposing effects: common or ubiquitous species will be added more rapidly than rare or localized species that have already been encountered and this will tend to reduce the diversity; at the same time, hitherto unrecorded species will be brought into the sample which will increase the diversity. When the two effects balance, the curve will level off. Suppose then that the curve becomes approximately horizontal for all k greater than some t . We may say that any number of quadrats in excess of t suffices to "represent" the population in the sense that enlargement of the sample would cause no further increase in diversity.

Provided levelling off occurs, $\mathcal{H}_z$, the estimate based on the largest sample (obtained by pooling all the z quadrats examined), could be taken as an estimate of H'_{pop} ; its sampling variance is unknown, however. (We write H'_{pop} rather than H_{pop} since what is estimated is the *average* diversity in the population.)

Alternatively, we may regard the sequence $\{\mathcal{H}_k\}$ for $k \geq t$ as a sequence of dependent estimates of H'_{pop} based on successively larger sample sizes. Denote the total number of individuals of all species in the first k combined quadrats by M_k . Then $\mathcal{H}_k$ and $\mathcal{H}_{k-1}$ are estimates of H'_{pop} based on totals of M_k and M_{k-1} individuals, respectively. Then $(M_k\mathcal{H}_k - M_{k-1}\mathcal{H}_{k-1})$ is the increment in *total* diversity that results from adding the $(M_k - M_{k-1})$ individuals in the k th quadrat to the first $(k-1)$ combined quadrats. Correspondingly,

$$(M_k\mathcal{H}_k - M_{k-1}\mathcal{H}_{k-1})/(M_k - M_{k-1}) \equiv h_k \quad (6)$$

is the increment in diversity *per individual*.

On the hypothesis that, when $k \geq t$, no change is expected in diversity as sample size is increased, it follows that $E(h_k) = H'_{\text{pop}}$. So as an estimator of H'_{pop} we may take the observed mean of the h 's; that is, we may put

$$\tilde{H}'_{\text{pop}} = \bar{h} = \frac{1}{(z-t+1)} \sum_{k=t}^z h_k \text{ and also } \widetilde{\text{var}}(\tilde{H}'_{\text{pop}}) = \text{var}(\bar{h}). \quad (7)$$

An entirely different argument leads to the same result. Baer (1953) has proved (and see Pielou, 1966a, b) that if (i) $M_k - M_{k-1} = 1$, and (ii), this single individual in the k th quadrat constitutes a random selection from all the individuals in the k combined quadrats, then the expected value of h_k

as defined in equation (6) is $E(h_k) = \mathcal{H}'_k$ (where the prime denotes use of the Shannon formula (2)). This result is exact. It will remain approximately true even when the conditions are not met, provided the quadrats are small and k is large. We have already hypothesized that $E(\mathcal{H}_k) = H'_{\text{pop}}$.

Although successive values of $\mathcal{H}_k$ are dependent, it is assumed that the h_k are independent. This is equivalent to assuming that, in removing quadrats one by one from the whole sample of z quadrats combined, we may treat the sampling as though it were done with replacement although it is done without. The assumption may safely be made provided a test shows no significant serial correlation between successive values of h_k .

Example

The method was tried on a population consisting of the ground vegetation in a 3000 sq. m tract of mixed woodland in Gatineau Park, near Hull, Quebec. The population consisted of all herbaceous vascular plants and all the woody plants and tree seedlings having a stem diameter at ground level not greater than 2 cm. The tract was sampled with 100 randomly-placed meter-square quadrats, of which only three were found empty. The sampling was done in late May, 1965, and was completed in six days to ensure that no appreciable changes in the weights of the plants would have time to occur during the course of the observations. For each quadrat the amount of each species was measured by cutting off the plants at ground level and weighing

TABLE 1
*The species, their weights in the total sample, and
the number of quadrats in which each occurred*

Species†	Weight (g)	Number of quadrats
<i>Acer saccharum</i> Marsh.	1650.0	92
<i>Erythronium americanum</i> Ker.	889.1	64
<i>Streptopus roseus</i> Michx.	879.7	48
<i>Aralia nudicaulis</i> L.	601.2	22
<i>Trillium grandiflorum</i> (Michx.) Salisb.	513.7	47
<i>Fagus grandifolia</i> Ehrh.	465.5	20
<i>Athyrium Filix-femina</i> (L.) Roth.	413.5	16
<i>Viburnum alnifolium</i> Marsh.	399.0	5
<i>Clintonia borealis</i> (Ait.) Raf.	354.4	4
<i>Tsuga canadensis</i> (L.) Carr.	341.2	6
<i>Lycopodium lucidulum</i> Michx.	318.7	1
<i>Carex</i> spp.	292.6	31
<i>Acer pensylvanicum</i> L.	250.3	3
<i>Aster acuminatus</i> Michx.	231.9	17
<i>Asarum canadense</i> L.	222.9	2
<i>Trillium erectum</i> L.	201.4	19

TABLE 1 (*continued*)

Species†	Weight (g)	Number of quadrats
<i>Dryopteris austriaca</i> (Jacq.) Woynar.	146.3	10
<i>Osmorhiza Claytoni</i> (Michx.) Clarke.	144.0	20
<i>Arisaema triphyllum</i> (L.) Schott.	117.6	24
<i>Circaea</i> spp.	116.5	9
<i>Sambucus pubens</i> Michx.	113.9	7
<i>Tilia americana</i> L.	109.8	5
<i>Fraxinus americana</i> L.	105.3	8
<i>Prenanthes altissima</i> L.	94.1	28
<i>Polygonatum pubescens</i> (Willd.) Pursh.	92.4	30
<i>Betula papyrifera</i> Marsh.	81.6	9
<i>Solidago caesia</i> L.	81.3	6
<i>Maianthemum canadense</i> Desf.	80.0	35
<i>Lonicera canadensis</i> Marsh.	74.6	1
<i>Pyrola elliptica</i> Nutt.	60.4	13
<i>Medeola virginiana</i> L.	58.4	6
<i>Pinus strobus</i> L.	54.6	7
<i>Uvularia grandiflora</i> Sm.	53.9	8
<i>Dryopteris marginalis</i> (L.) Gray.	51.4	1
<i>Actaea alba</i> (L.) Mill.	46.5	4
<i>Viola</i> spp.	41.1	33
<i>Cornus alternifolia</i> L. f.	37.9	3
<i>Populus balsamifera</i> L.	33.6	6
<i>Smilacina racemosa</i> (L.) Desf.	29.4	4
<i>Prunus serotina</i> Ehrh.	19.3	12
<i>Acer rubrum</i> L.	18.2	1
<i>Dentaria diphylla</i> Michx.	17.9	3
<i>Aster cordifolius</i> L.	11.9	2
<i>Botrychium virginianum</i> (L.) Sw.	11.6	5
<i>Galium triflorum</i> Michx.	11.2	3
<i>Rubus</i> sp.	9.6	1
<i>Populus grandidentata</i> Michx.	8.7	1
<i>Sanicula</i> spp.	8.4	4
<i>Onoclea sensibilis</i> L.	7.2	1
<i>Aralia racemosa</i> L.	7.0	1
<i>Actaea rubra</i> (Ait.) Willd.	5.3	1
<i>Brachyelytrum erectum</i> (Schreb.) Beauv.	5.2	3
<i>Tiarella cordifolia</i> L.	3.9	2
<i>Quercus borealis</i> Michx. f.	3.7	3
<i>Caulophyllum thalictroides</i> (L.) Michx.	3.6	1
<i>Ranunculus abortivus</i> L.	2.5	1
<i>Carya cordiformis</i> (Wang.) K. Koch.	2.1	2
<i>Equisetum hyemale</i> L.	1.5	1
<i>Veronica officinalis</i> L.	1.4	2
<i>Trientalis borealis</i> Raf.	1.3	1
<i>Mitella diphylla</i> L.	1.0	2
<i>Mitchella repens</i> L.	0.6	1

† Nomenclature follows Gleason & Cronquist (1963).

them (fresh) to the nearest decigram. N_i was then the number of decigrams (an integer) of species i and these N_i 's were substituted in equation (1) to determine H for each quadrat.

Table 1 gives a summary of the results. For each species is given: (i) its total weight in the whole sample of 100 quadrats; and (ii) the number of quadrats in which it was found. For four genera the identification could not be carried to the species level with the young non-flowering specimens available; these were *Carex* and *Viola* (both with several species); *Circaea* (*C. alpina* L. and *C. quadrisulcata* (Maxim.) Franch. & Sav.); and *Sanicula* (*S. marilandica* L. and *S. gregaria* Bickn.). Treating these genera as "species", a total of 62 species was found in the sample. Besides these, five other species that did not occur in any of the quadrats were found elsewhere in the tract. These were: *Juglans cinerea* L., *Geum canadense* Jacq., *Thalictrum polygamum* Muhl., *Polystichum acrostichoides* (Michx.) Schott., and *Gymnocarpium Dryopteris* (L.) Newm. It is believed that the list of 67 species for the tract is exhaustive.

3. Results

(i) The number of species per quadrat ranged from 0 to 14, with a mean of 7.27.

(ii) The weight of plant material per quadrat ranged from 0 to 359.3 g, with a mean of 100.13 g.

(iii) Excluding the three empty quadrats (for which diversity is undefined), the diversities of the separate quadrats, H , ranged from 0 (for the two single-species quadrats) to 2.096 natural bels. The mean was $\bar{H} = 1.227$ natural bels. The histogram of Fig. 1 shows the observed distribution of H .

(iv) The weight of plant material in a quadrat and its diversity were uncorrelated.

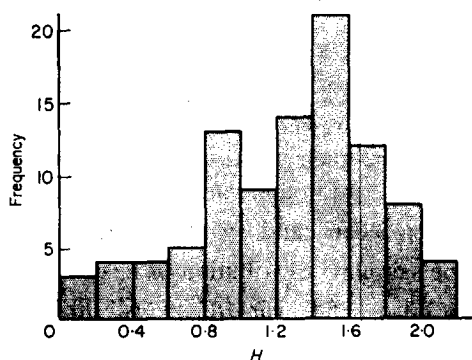


FIG. 1. The distribution of H , the diversity of a single quadrat, for 97 quadrats.

(v) In Fig. 2 are shown the curves of $\mathcal{H}_k$ versus k (the number of pooled quadrats) and versus the accumulated weight of plant material in the pooled quadrats. (If the plant weight were the same in all quadrats the two curves would be identical; to avoid overlap of the two curves the ordinate scales have been displaced with respect to each other). From inspection of these curves a value for t of 70 was chosen; i.e. it was assumed that 70 quadrats were representative of the population in that no increase in diversity was to be expected on enlargement of the sample. Values of h_k for $k = 70, 71, \dots, 100$ were calculated as described above. The 74th and 90th quadrats were

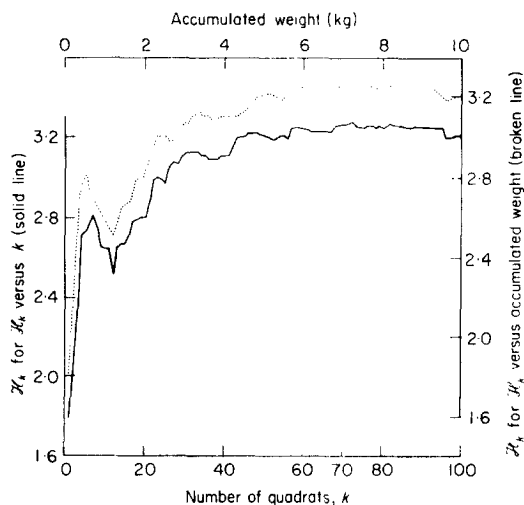


FIG. 2. $\mathcal{H}_k$ versus k (solid line, left-hand ordinate scale); and $\mathcal{H}_k$ versus accumulated weight of plant material (broken line, right-hand ordinate scale).

empty; for such quadrats h is undefined, since both numerator and denominator in equation (6) are identically zero. There were, therefore, 29 values of the variate h and their serial correlation was insignificant. With natural bells as the unit the mean and variance of the observed values were:

$$\bar{h} = 3.056 \quad \text{and} \quad \text{var}(h) = 0.7638$$

whence, assuming the sampling distribution of h_k to be normal we have

$$\bar{H}'_{\text{pop}} = 3.056 \quad \text{and} \quad \widehat{\text{var}}(\bar{H}'_{\text{pop}}) = \widehat{\text{var}}(\bar{h}) = 0.026,$$

so that the 95% confidence limits for the population diversity were

$$2.74 \leq H'_{\text{pop}} \leq 3.37.$$

4. The Evenness Component of Diversity

For a given number of species, the diversity of a collection is a maximum when the individuals are distributed among the species as evenly as possible. Conversely, if only a few of the species contain a large majority of the individuals, the diversity is low. As a measure of the evenness with which the individuals are divided among the species in any collection we may take the ratio of the observed diversity to the maximum possible for the same number of species (Pielou, 1966*a*, *b*).

For Type *A* collections, the maximum possible diversity is

$$H_{\max} = \frac{1}{N} \log \frac{N!}{\{[N/s]!\}^{s-r} \{([N/s] + 1)!\}^r}, \quad (8)$$

where $[N/s]$ is the integer part of N/s and $r = N - s[N/s]$. The population value of the collection's evenness is then $J = H/H_{\max}$.

For all other collections the estimation of evenness requires that s , the number of species, be known or estimable. We shall not consider here possible methods of estimating s when it is unknown (but see Good (1956)). When it is known, the greatest possible value of the diversity may be calculated since

$$H'_{\max} = - \sum \frac{1}{s} \log \frac{1}{s} = \log s$$

(Shannon's formula is used because we assume the population to be indefinitely large). Then for large collections (Types *B*, *C*, *D* and *E*) the population evenness is given by

$$J' = H'/\log s \text{ with estimator } \hat{J}' = \hat{H}'/\log s,$$

where the same units are used in numerator and denominator. Clearly, if s is taken as the base of the logarithms in calculating diversity, the denominator becomes $\log_e s = 1$ and $J' = H'$ identically.

For the plant community described in section 3 the estimated evenness of the population is

$$J'_{\text{pop}} = \frac{\hat{H}'_{\text{pop}}}{\log s} = \frac{3.056}{\ln 67} = 0.727.$$

It is also interesting to consider the distribution of J , the evenness within each quadrat, and the correlation between single-quadrat values of H and J . The observed distribution of J is shown in Fig. 3 for the 95 quadrats containing two or more species (for one-species quadrats, J is undefined); the mean is $\bar{J} = 0.657$. From the scatter diagram in Fig. 4 it is seen that J and H are correlated and, in particular, that for no quadrat was H large and J small. This presumably stems from the fact that for a quadrat to have these

properties it would have to contain a large number of species and thus a high value of $H_{\max}$. In patchy, coarse-patterned vegetation, this is unlikely to happen.

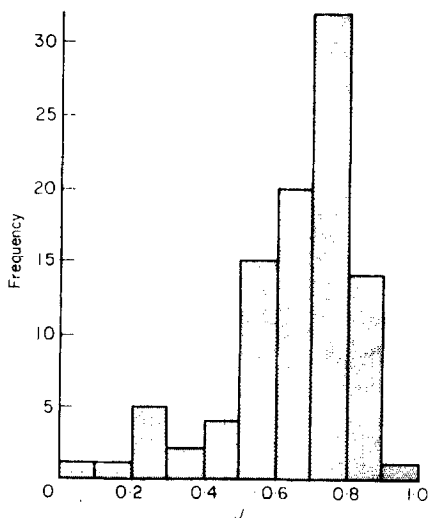


FIG. 3. The distribution of J , the evenness of a single quadrat, for 95 quadrats.

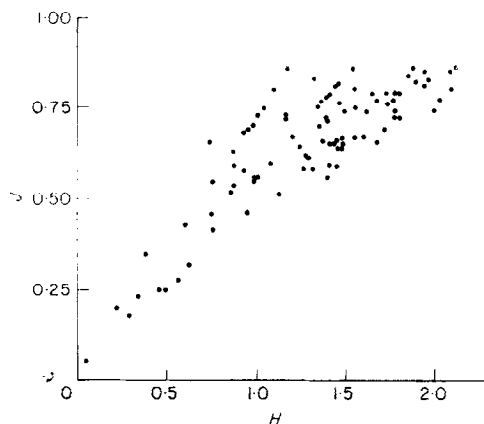


FIG. 4. The relation between single-quadrat values of H and J .

5. Glossary of Symbols

The letter H in various forms denotes the diversity *per individual* in a collection. Thus:

H is the population value of the diversity of any collection all of whose members have been identified and counted. It is calculated from Brillouin's formula (1), which requires a knowledge of the actual numbers of individuals of all species;

H' is the population value of the average diversity, defined by Shannon's formula (2), in an indefinitely large population. It is based on the proportions of the various species in a population of unknown size. It can never be directly calculated but must always be estimated;

$\hat{H}$ and $\hat{H}'$ are estimators of H' ;

H_{pop} is the population diversity, defined by Brillouin's formula, in a sessile population. The subscript emphasizes the fact that H_{pop} necessarily exceeds the expected value of H for a quadrat within the population since the contents of a quadrat are a non-random sample of the population;

H'_{pop} is the average population diversity in an indefinitely large sessile population, defined by Shannon's formula;

$\hat{H}'_{\text{pop}}$ is an estimator of H'_{pop} ;

$\mathcal{H}_k$ is the diversity (from Brillouin's formula) of the pooled contents of k quadrats placed at random in a sessile population;

$\mathcal{H}'_k$ is the same calculated from Shannon's formula, with $p_i = N_i/N$;

h_k is the increment in diversity (per added individual) that results from adding the individuals in a new quadrat to the pooled contents of $k-1$ quadrats. It is not directly measurable but is calculable from equation (6).

The letter J denotes evenness:

$J = H/H_{\text{max}}$ has Brillouin's formula in both numerator and denominator.

H_{max} is defined in equation (8);

$J' = H'/H'_{\text{max}}$ has Shannon's formula in both numerator and denominator.

$H'_{\text{max}} = \log s$, where s is the number of species;

$J' = \hat{H}'/H'_{\text{max}}$ is an estimator of the evenness in an indefinitely large population;

$J'_{\text{pop}} = \hat{H}'_{\text{pop}}/H'_{\text{max}}$ is the same in an indefinitely large sessile population.

REFERENCES

- BASHARIN, G. P. (1959). *Theory Probab. Applic.* 4, 333.
 BAER, R. M. (1953). "Information Theory in Biology" (H. Quastler, ed.). Urbana: University of Illinois Press.
 BRILLOUIN, L. (1960). "Science and Information Theory". (2nd edn.) New York: Academic Press.

- GLEASON, H. A. & CRONQUIST, A. (1963). "Manual of Vascular Plants of Northeastern United States and Adjacent Canada". Princeton: D. Van Nostrand Co. Inc.
- GOLDMAN, S. (1953). "Information Theory in Biology" (H. Quastler, ed.). Urbana: University of Illinois Press.
- GOOD, I. J. (1950). "Probability and the Weighing of Evidence". London: Charles Griffith & Co. Ltd.
- GOOD, I. J. (1953). *Biometrika*, **40**, 237.
- GOOD, I. J. (1956). *Biometrika*, **43**, 45.
- HAIRSTON, N. G. (1959). *Ecology*, **40**, 404.
- LLOYD, M. (1964). *Am. Nat.* **98**, 190.
- LLOYD, M. & GHELARDI, R. J. (1964). *J. Anim. Ecol.* **33**, 217.
- MACARTHUR, R. H. (1964). *Am. Nat.* **98**, 387.
- MACARTHUR, R. H. (1965). *Biol. Rev.* **40**, 510.
- MACARTHUR, R. H. & MACARTHUR, J. W. (1961). *Ecology*, **42**, 594.
- MARGALEF, D. R. (1958). *Gen. Syst.* **3**, 36.
- PIELOU, E. C. (1966a). *J. Theoret. Biol.* **10**, 370.
- PIELOU, E. C. (1966b). *Proc. Fifth Berkeley Symp. math. Statist. Probab.* (In press).
- SHANNON, C. E. & WEAVER, W. (1963). "The Mathematical Theory of Communication". Urbana: University of Illinois Press.